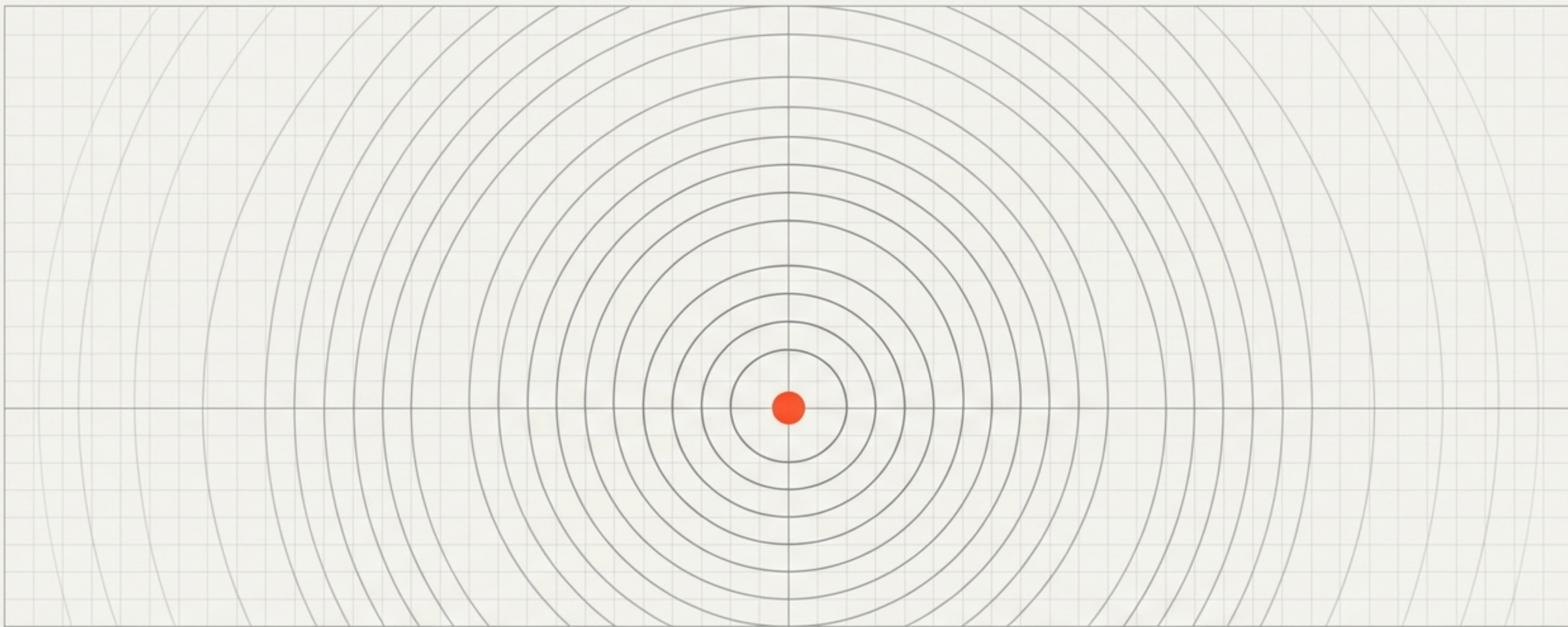




Diffusion, Branch Number & Avalanche

The mathematics of cryptographic spreading.

The Macro Goal: The Crack in the Ice

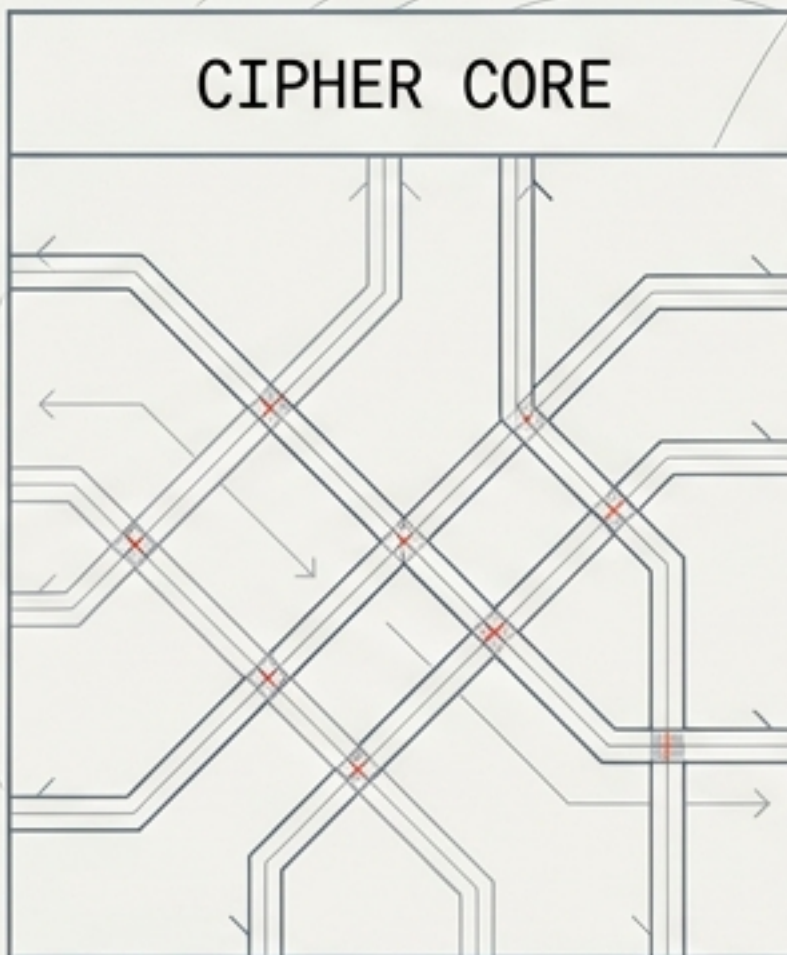
In cryptography, diffusion is the art of making a tiny crack spread across the entire pond. A single changed bit must ripple outward until the state is affected in ways that are impossible to predict without the key.

INPUT STATE (4x4)

0	0	0	0
0	0	1	0
0	0	0	0
0	0	0	0



CIPHER CORE



OUTPUT STATE (4x4)

0	1	0	1
1	0	1	1
0	1	1	0
1	0	0	1

DIFFUSED OUTPUT: ~50% BITS CHANGED

The Twin Pillars: Confusion vs. Diffusion

Confusion

Meaning: Makes relationships complicated and nonlinear. Bends patterns.

Mechanism: S-boxes, inverses, nonlinear operations.

Vulnerability: Without diffusion, **changes stay completely localized.**

Diffusion

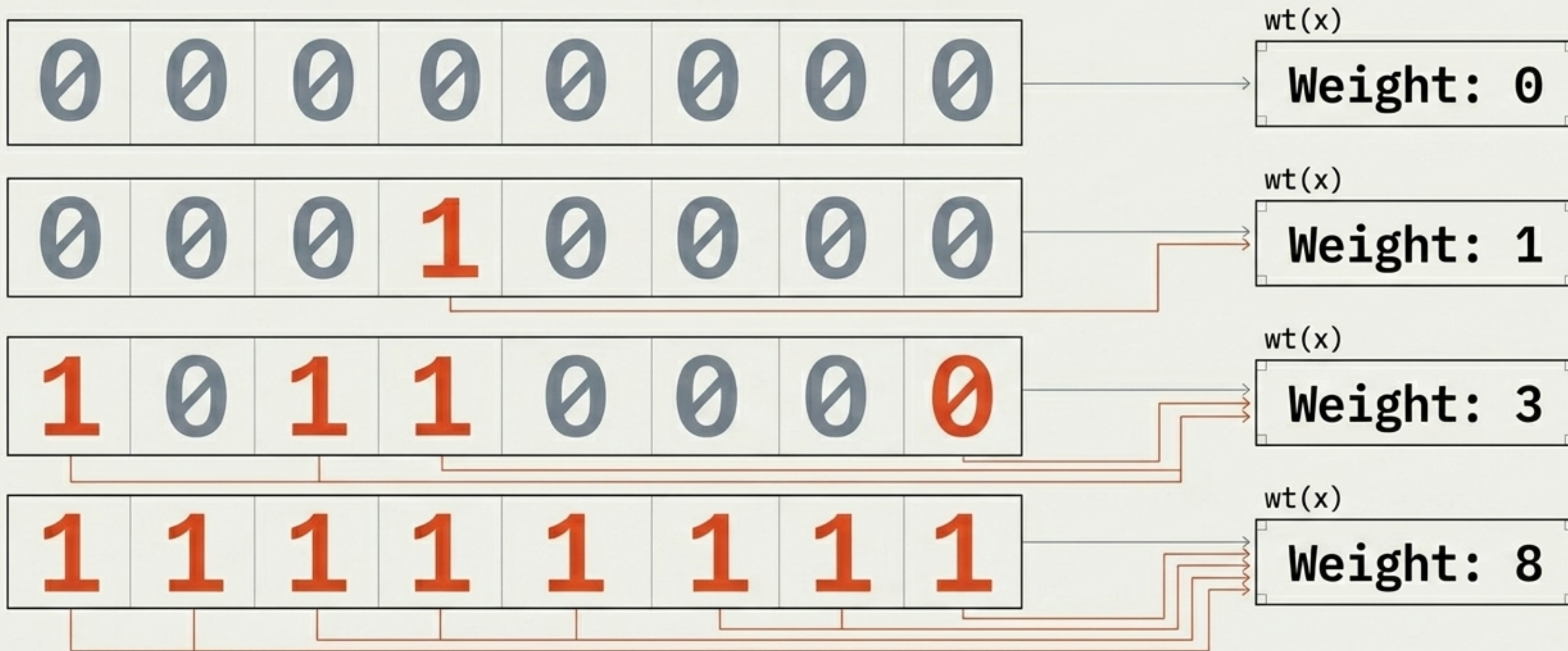
Meaning: Spreads local changes across many positions. Carries bent patterns everywhere.

Mechanism: Permutations, matrices, linear layers (e.g., MixColumns).

Vulnerability: Without confusion, **relationships remain algebraically predictable.**

The Measuring Tape: Hamming Weight

Hamming weight, written as $wt(x)$, is simply the number of active bits (1s) in a single string.



Comparing States: Hamming Distance

To find the distance between two strings, XOR them and count the 1s.

Hamming distance(a, b) = wt(a XOR b)

String A: 10110010

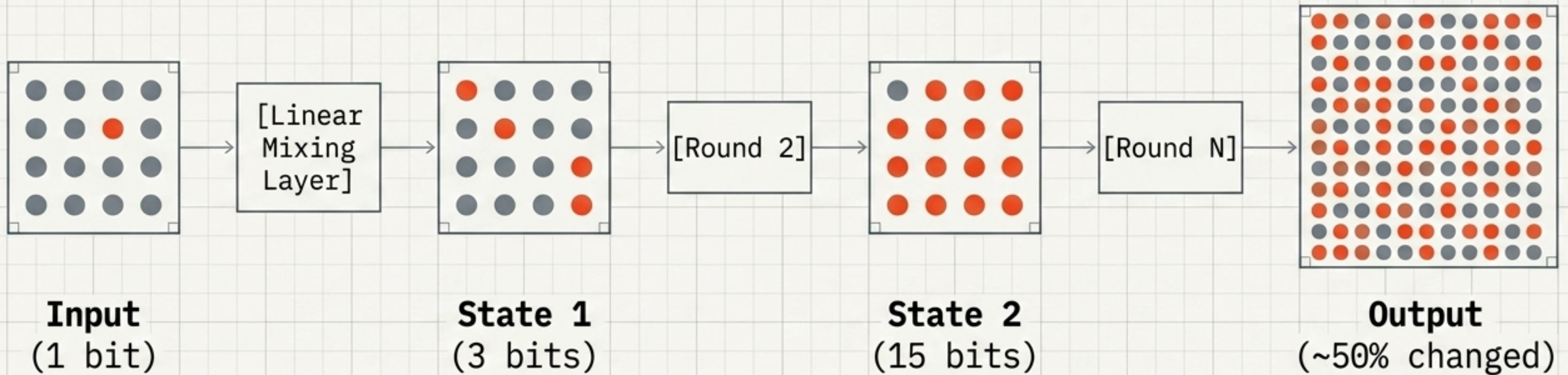
String B: 10110111

a XOR b: 00000101

Distance = 2

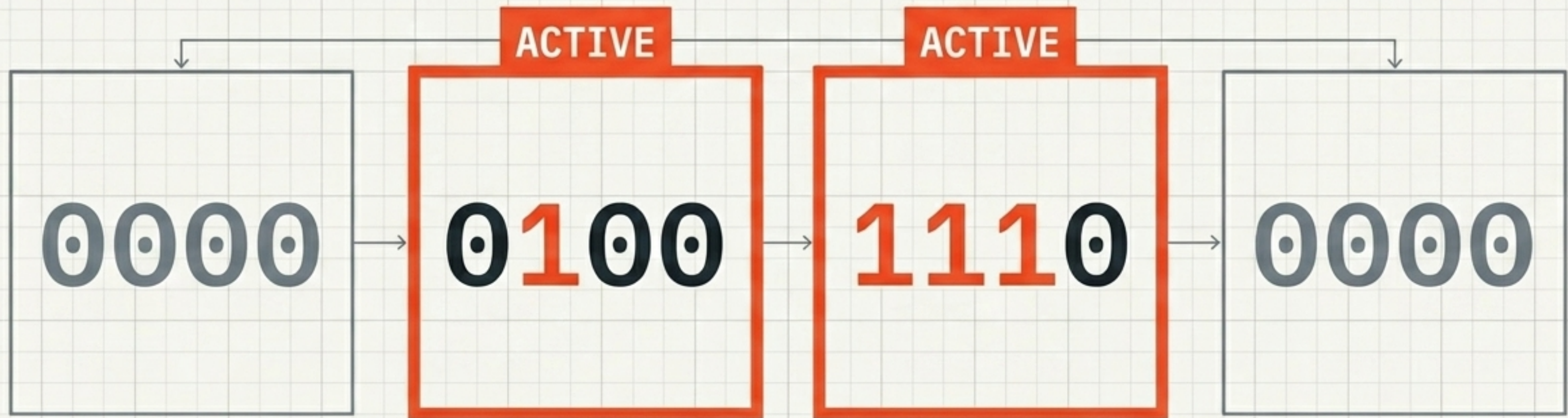
The Avalanche Effect: Visualizing the Ripple

Avalanche is a multi-round behavior. A single small input change amplifies mathematically through the round structure until roughly half of all output bits are flipped.



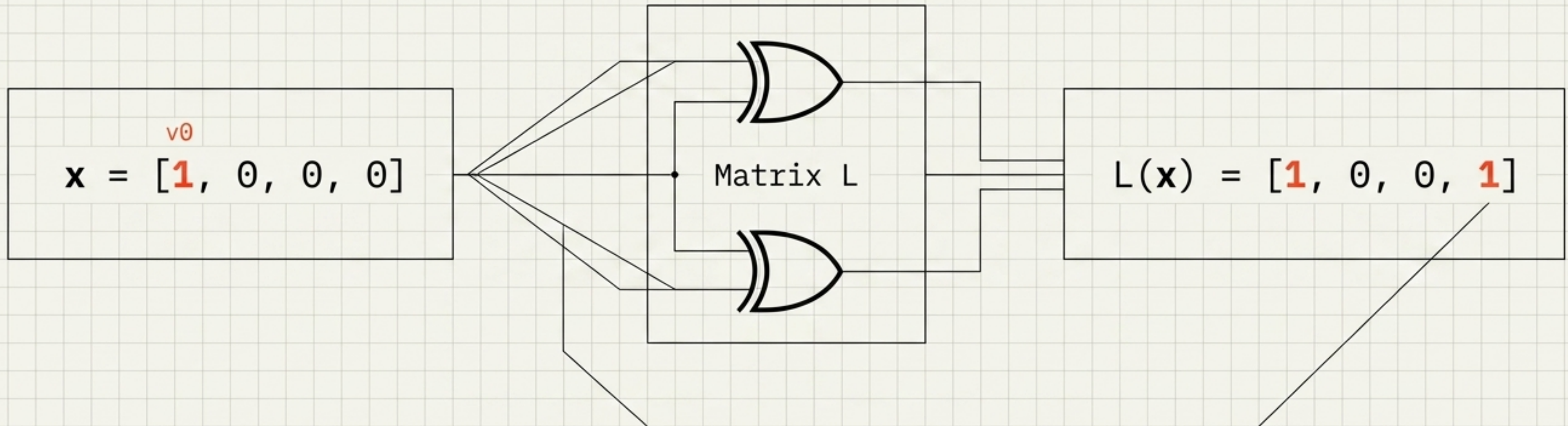
Tracking Chunks: Active Components

Insight: Cipher analysis rarely tracks every single bit. Instead, we track active components. If a unit (a nibble, byte, or S-box input) contains even one nonzero difference, the entire component is flagged as ACTIVE.



The Mechanism: A Linear Layer in Action

1 active input bit is mathematically forced to become 2 active output bits. Diffusion has begun.



$$L(v_0, v_1, v_2, v_3) = (v_0 \text{ XOR } v_1, v_1 \text{ XOR } v_2, v_2 \text{ XOR } v_3, v_0 \text{ XOR } v_3)$$

The Benchmark: Defining Branch Number

Ignore zero differences.
We only measure when
there is an active input.

The worst-case scenario.
Attackers look for weak trails, so
we score the layer by the smallest
possible activity it can produce.

$$B = \min_{\substack{x \neq 0}} (\underbrace{wt(x)} + \underbrace{wt(L(x))})$$

Total activity: The
input weight plus the
output weight.

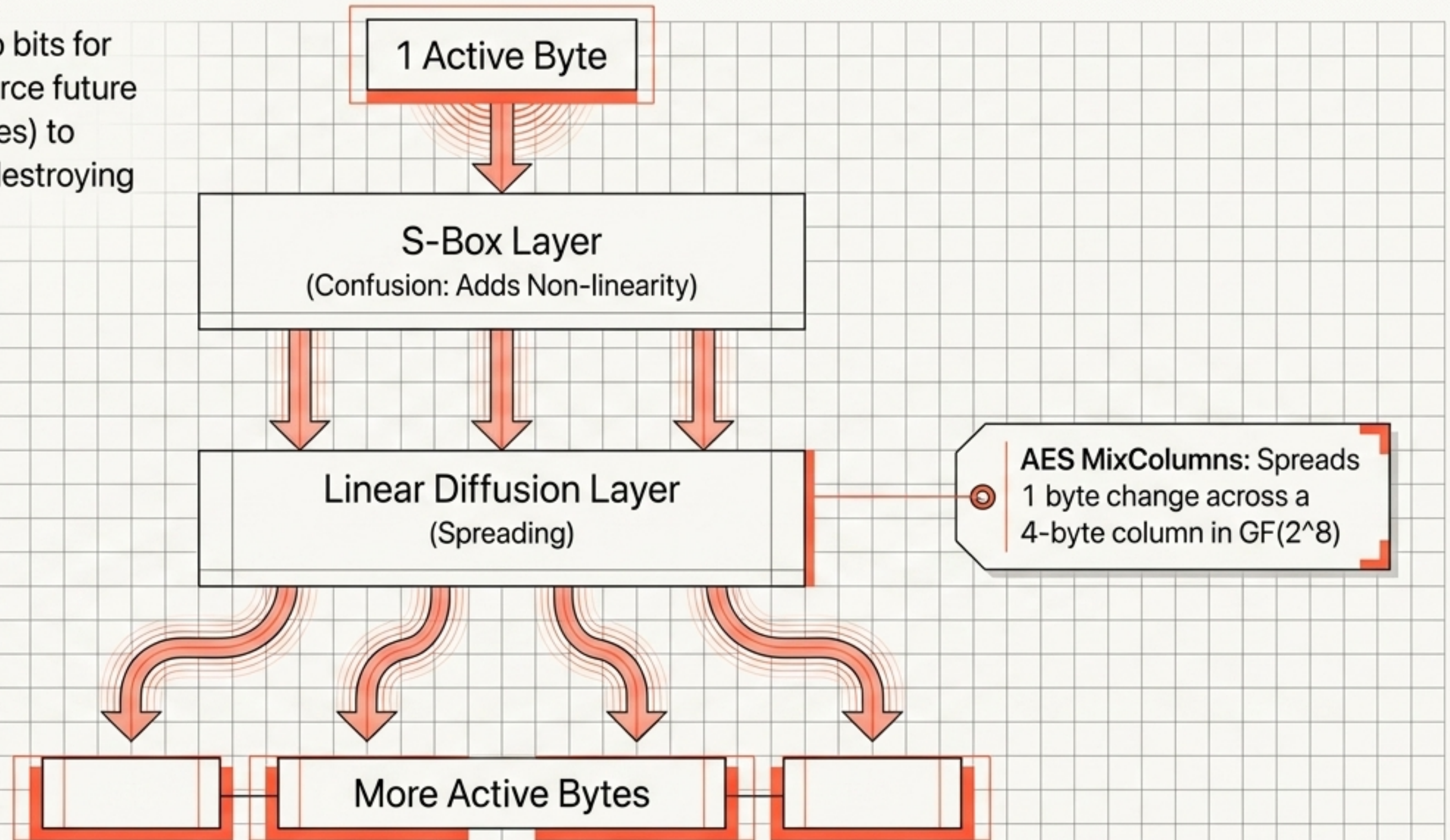
Stress-Testing the Layer

No matter which nonzero input combination you inject, this specific linear layer mathematically guarantees at least 3 active bits in total.

Layer Stress-Test				
Input	$\text{wt}(\mathbf{x})$	Output	$\text{wt}(L(\mathbf{x}))$	Total
1 0 0 0	1	1 0 0 1	2	3
1 1 0 0	2	0 1 0 1	2	4
1 0 1 0	2	1 1 1 1	4	6
SYSTEM FLOOR: $B = 3$				

Synthesis: The Wide-Trail Strategy

Good diffusion doesn't just flip bits for decoration. It is designed to force future nonlinear components (S-boxes) to become active, permanently destroying simple attack trails.



Diagnostic Checklist: 4 Common Misconceptions

Avalanche is instant. -> FALSE.

Avalanche is a multi-round behavior; individual components only begin the process.

Weight = Distance. -> FALSE.

Weight counts 1s in a single string.
Distance counts differences between two strings.

**Judge by average diffusion.
-> FALSE.**

Worst-case behavior matters most. Branch number scores the smallest possible activity.

**Linear layers are useless alone.
-> FALSE.**

They are essential for diffusion; they just perform a different job than S-boxes.

Executive Summary / System Architecture

Core Formulas

Hamming Distance:

$$\text{Distance}(a, b) = \text{wt}(a \text{ XOR } b)$$

Branch Number:

$$B = \min_{x \neq 0} (\text{wt}(x) + \text{wt}(L(x)))$$

Core Concepts

Weight: How many 1s are in this string?

Distance: How many positions changed?

Avalanche: Does one bit flip eventually change many output bits?

Branch Number: What is the worst-case spreading limit?

Active Component: Is this chunk affected by a difference?